The Emergence of Relational Time (Page-Wootters Mechanism)

1. Global state

no external time

$$|\Psi\rangle$$

$\hat{H}|\Psi\rangle = 0$ (static Wheeler-DeWitt block)

The field state carries correlations, not a time parameter.

2. Decomposition

clock-system split

$$|\Psi\rangle = \sum_n |n\rangle_C \otimes |\psi(n)\rangle_S$$



clock ladder: clock-basis label n

3. Conditional readout

relative to the clock

$\langle 0|_C\Psi\rangle \; |\psi(0)\rangle_S$

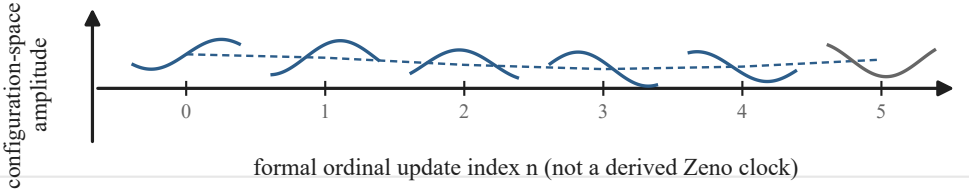
$\langle 1|_C\Psi\rangle \; |\psi(1)\rangle_S$

$\langle 2|_C\Psi\rangle \; |\psi(2)\rangle_S$

$\langle 3|_C\Psi\rangle \; |\psi(3)\rangle_S$

4. Reconstructed sequence

ordinal dynamics



formal ordinal update index n (not a derived Zeno clock)

Dynamics are recovered as conditional correlations $\{|\psi(n)\rangle_S\}$ inside the frozen global state.